Scientific Summary

Phylogenetic Analysis of the Hemoglobin Beta (HBB) Gene

The Hemoglobin Beta gene is a highly conserved protein-coding gene that codes for the beta-globin subunit of adult hemoglobin. Due to its fundamental physiological role, which is playing part in transporting oxygen through blood to organs, it is extensively studied to investigate evolutionary relationships and functional adaptation.

Phylogenetic analyses of HBB sequences consistently demonstrate that the gene is highly conserved across mammals**,** showing that its structure and function has remained preserved. Mammalian HBB genes show generally high nucleotide and amino acid sequence identity. The conserved coding regions maintain critical amino acid residues that cause heme binding, αβ-globin interactions, and oxygen binding. Changes in these amino acid residues can lead to unstable hemoglobin molecules which may lead to anemia and reduced oxygen transport efficiency.

In this project, the **human HBB protein sequence (RefSeq Accession: NP_000509.1)** serves as the reference sequence for identifying homologous HBB proteins from multiple mammalian species using **BLASTp.** The retrieved protein sequences are aligned using the **ClustalW** algorithm in **MEGA 12** to identify conserved amino acid residues and functionally important regions. The aligned sequences are subsequently used to construct a **Neighbor-Joining phylogenetic tree**, providing insights into the evolutionary relationships among the HBB proteins across different species. The resulting phylogenetic tree is further visualized and annotated using **Interactive Tree of Life (iTOL)** to improve interpretation and presentation.

This project demonstrates a complete comparative bioinformatics workflow, integrating sequence retrieval, homology searching, multiple sequence alignment, phylogenetic reconstruction, and tree visualization. Through thee analyses, participants gain practical experience in evolutionary bioinformatics while developing an understanding of how sequence conservation reflects the functional importance of proteins. The study also highlights the value of phylogenetic analysis in exploring molecular evolution, comparative genomics, and the conservation of essential genes and proteins across species.